

MOHAMMED MAHYOUB

Senior ITS & Systems Integration Engineer | Applied AI & Computer Vision

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PROFESSIONAL SUMMARY

Senior Service Engineer with 10+ years of multidisciplinary engineering experience, including 6+ years supporting intelligent transportation, smart infrastructure, and traffic enforcement systems across the UAE. I integrate cameras, LiDAR, sensors, control units, networks, databases, and backend platforms into reliable end-to-end operational systems, and lead FAT/SAT, commissioning, calibration, and structured root-cause analysis. Currently pursuing an MSc in Artificial Intelligence at UWE Bristol, applying machine learning and computer vision to smart mobility, with a focus on data quality, leakage-controlled evaluation, efficient neural networks, and explainability.

CORE EXPERTISE

ITS & Traffic Enforcement | Systems Integration | FAT / SAT, Commissioning & Calibration | Machine Vision & LiDAR | Root-Cause Analysis | Networking (Cisco, MikroTik, MPLS, VoIP/IMS) | Virtualisation & Data Centre (VMware ESXi, iSCSI) | Embedded Systems & Automation | Python, PyTorch & Machine Learning

PROFESSIONAL EXPERIENCE

Senior Service Engineer – Smart Infrastructure & ITS Systems | VITRONIC Machine Vision

Oct 2019 – Present | United Arab Emirates

- Support an installed base of 800+ intelligent transportation and traffic enforcement systems across the UAE.
- Integrate cameras, LiDAR, sensors, control units, networks, databases, and backend platforms into reliable end-to-end systems.
- Execute FAT, SAT, commissioning, calibration, interface validation, acceptance testing, and performance verification.
- Diagnose complex hardware, software, communication, database, and system-interface faults using structured root-cause analysis.
- Coordinate incident response, preventive maintenance, upgrades, and configuration changes with customers, vendors, and engineering teams.
- Prepare technical reports, test records, incident analyses, and system handover documentation.

Electronics, Embedded Systems & Automation Engineer | Electronic Science Company

Jun 2015 – Aug 2019 | Al Hudaydah, Yemen

- Designed, assembled, tested, and troubleshoot electronic circuits, embedded control systems, and sensor-based automation.
- Programmed PIC, AVR, and Arduino microcontrollers; integrated sensors, displays, communication modules, controllers, and PLCs.
- Performed PCB prototyping, hardware debugging, functional testing, and technical validation.
- Developed VB.NET desktop applications for monitoring, diagnostics, reporting, and workflow automation, with deployment and customer support.

Independent Engineering Consultant – Network Infrastructure & Technical Projects | Independent Engineering Projects

Mar 2012 – May 2018 | Part-time / Project-based

- Designed and supported wired and wireless infrastructure using routing, switching, firewall, VPN, MikroTik, and Ubiquiti, including coverage and point-to-point link planning.
- Designed, built, and validated MPLS, OSPF, BGP, VRF, and traffic-engineering architectures using Cisco platforms and GNS3.
- Designed and deployed photovoltaic power systems with battery storage and inverters, including solar supply for remote telecommunications sites.
- Developed engineering tools for RF calculations and reporting; prepared network diagrams, configurations, and technical documentation.

SELECTED PROJECTS

- **Efficient Neural Networks for Vehicle Type Classification (VTID2):** MSc research in PyTorch — content audit of 4,356 images (2,425 exact duplicates), leakage-controlled evaluation of MLP, CNN, depthwise-separable and SE CNNs, efficiency analysis, and Grad-CAM.
- **Vehicle Type Classification Using NGSIM US-101:** vehicle-level, leakage-aware ML with Logistic Regression, Decision Tree and Random Forest; archived on Zenodo (DOI 10.5281/zenodo.23033235).

- **Autonomous Quadcopter UAV:** ArduPilot APM 2.6 build with MAVLink telemetry, GPS waypoint missions, failsafes and 5.8 GHz FPV; flight-tested across autonomous modes with log-based root-cause fixes.
- **IMS Multimedia Services — VoIP, IPTV & VoLTE:** Cisco CUCM/CME with H.323/SIP interworking, IPTV streaming, and VoLTE performance analysis over an IMS core.
- **Software-Defined Data Center (SDDC):** VMware ESXi with FreeNAS/ZFS storage over iSCSI, vSwitch networking, and AD, Citrix XenApp, web and Exchange 2013 services.
- **Network Infrastructure Design:** multi-site wireless/RF links (MikroTik, Ubiquiti) and a carrier-grade MPLS backbone (OSPF, MPLS-TE, VRF, MP-BGP).
- **Embedded Systems & Monitoring Applications:** PIC16 firmware (ADC, EEPROM, I2C RTC, keypad/LCD access control), PLC monitoring, and a VB.NET monitoring & diagnostics console with RF calculators.
- **Solar Energy Systems & RFID Library Automation:** PV design, sizing and deployment for remote telecom sites; ISO 15693 (13.56 MHz) RFID system for library check-in/out, security and inventory.

EDUCATION

MSc Artificial Intelligence | University of the West of England (UWE Bristol)

May 2026 – Jun 2028 | In progress — machine learning, neural networks, computer vision, model evaluation and efficiency

BEng Electrical Engineering – Communications & Electronics | Sana'a University

Sep 2012 – Jun 2017 | UAE Ministry of Education degree equivalency obtained

CERTIFICATIONS & RECOGNITION

- Certificate of Appreciation — Dubai Police HQ, UAE (2023)
- Cisco Certified Network Associate — Routing & Switching (CCNA) • CompTIA A+
- PIC Microcontroller Programming — mikroC & Proteus (training)

TECHNOLOGIES & LANGUAGES

Technologies: Python, PyTorch, scikit-learn, Pandas, VB.NET, mikroC, PIC/AVR/Arduino, PLC, ArduPilot/MAVLink, Cisco, MikroTik, Ubiquiti, GNS3, OPNET, Cisco CUCM/CME, VMware ESXi, FreeNAS/iSCSI, Citrix XenApp, Exchange, Active Directory.

Languages: Arabic (Native / Bilingual) | English (Professional Working Proficiency)